Computer Vision

HW 3A: Image Classification using CNN

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Introduction

Image classification is an important application of computer vision as it has many applications for automating the recognition of visual information. From autonomous vehicles, to robotics, to medical applications, there are many very important ways that image classification can be used for. This paper will dive into training and using convolutional neural networks (CNN) to classify objects in images, and analyze how layers and number of classes affects the accuracies of the CNNs.

Experiments and Results

	Net1	Net2
Plane	64.7%	73.5%
Car	83.2%	82.0%
Bird	47.9%	57.9%
Cat	21.9%	57.1%
Deer	59.8%	53.7%
Dog	56.2%	57.3%
Frog	77.5%	85.4%
Horse	57.2%	73.8%
Ship	76.9%	87.7%
Truck	47.1%	71.0%

Average	59.24%	69.94%
Running time	31.57 seconds	35.65 seconds

Net1 Confusion Matrix

X/1000	Plane	Car	Bird	Cat	Deer	Dog	Frog	Horse	Ship	Truck
Plane	647	46	51	8	35	18	16	10	147	22
Car	15	832	11	6	9	8	14	4	65	36
Bird	72	12	479	32	142	80	124	19	33	7
Cat	36	35	107	219	112	275	115	36	48	17
Deer	26	8	84	25	598	43	148	47	20	1
Dog	18	11	116	82	79	562	59	51	16	6
Frog	4	13	48	49	58	31	775	3	18	1
Horse	13	9	74	33	147	108	25	572	8	11
Ship	69	71	20	5	14	18	12	5	769	17
Truck	37	247	19	16	13	25	25	28	119	471

Net2 Confusion Matrix

X/1000	Plane	Car	Bird	Cat	Deer	Dog	Frog	Horse	Ship	Truck
Plane	735	17	23	31	16	9	12	11	126	20
Car	23	820	6	9	2	8	18	4	68	42
Bird	81	4	579	87	36	52	96	30	28	7
Cat	25	10	80	571	46	131	83	22	25	7
Deer	34	2	101	86	537	22	107	86	21	4
Dog	19	3	49	232	30	573	36	45	8	5

Frog	18	4	32	51	6	16	854	5	12	2
Horse	26	3	31	76	47	50	19	738	7	3
Ship	54	9	10	14	4	5	6	0	877	21
Truck	50	90	6	32	6	5	17	7	77	710

	Net3
Deer	62.4%
Dog	49.7%
Horse	75.0%
Other	93.8%
Average	84.38%
Running time	35.46 seconds

Net3 Confusion Matrix

	Horse	Dog	Deer	Other
Horse	750/1000	36/1000	70/1000	144/1000
Dog	83/1000	497/1000	44/1000	376/1000
Deer	62/1000	14/1000	624/1000	300/1000
Other	113/7000	158/7000	162/7000	6567/7000



Discussion

Shown in the results from the neural networks, there is an obvious pattern showing that the network is actually learning based on the data. For each of the testing data results, the majority of the labels had it so that the correct label had the highest score out of all of them. This shows that it is not randomly guessing the correct label most of the time, but that the network is actually learning. There is an exception for this, for example with Net1, the cat had 219 correct labels but also 275 incorrect ones labeled as dog. This makes sense that these two are mixed up as they can have similar body shapes and move in similar ways. In the Net2 network, deer had the worst accuracy out of all the classes and it was mistaken the most for the frog class. Though these animals don't have much in common physically, an explanation for why these might be getting confused the most is the environment around them. Since deer and frogs are found in green luscious areas, the network could be learning based on the background color and texture, confusing those two animals.

When looking at the accuracies of the neural networks, the Net3 network had the highest accuracy by far, which can be attributed to the fact that it has much less classes to classify. Since it just has to identify 3 of the classes, and the rest that are not in those classes, there is a much higher chance that the image is one of the other 7 images, allowing for the other class to score an exceptionally high average. Other than that Net2 scored higher on average than Net1. The reason that this is true is because Net2 had many more output channels than Net1 and Net2 had an additional convolutional layer, allowing for Net2 to learn more features at each level.

Analyzing the misclassified images from the Net2 network, based purely on the visualization of the images, some conclusions could be made for why the network misclassified

the images. For example with the first image, the car doors are open, which kind of imitate the wings of a bird, therefore it was classified as so. With the deer image, since the surroundings for it is completely blue, the network likely classified it as a plane due to planes always being in the sky, and therefore surrounded by blue. Similar reason for why the bird label was classified as a frog, as most frogs are in a green environment. Finally, the truck was likely misclassified as a car just solely due to cars having similar shapes and properties as some trucks, so it would be easy for them to get mixed up.

Works Cited

"Training a Classifier." *PyTorch Tutorials*, PyTorch, 30 Sept. 2025, docs.pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html.